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United States Patent Application Publication	20250281205
Kind Code	A1
Publication Date	September 11, 2025
Inventor(s)	Bales; Joel

SURGICAL SYSTEM AND METHODS OF USE

Abstract

A spine rod connector includes a first portion having end surfaces and side surfaces. The first portion includes top and bottom surfaces. The first portion includes a first aperture extending between and through the end surfaces and spaced apart holes extending through the top surface such that the holes are in communication with the first aperture. The spine rod connector includes a second portion having end surfaces and side surfaces. The second portion includes top and bottom surfaces. The second portion includes a second aperture extending between and through the side surfaces of the second portion and spaced apart holes extending through the top surface of the second portion such that the holes of the second portion are in communication with the second aperture.

Inventors: **Bales; Joel (Hernando, MS)**

Applicant: **Warsaw Orthopedic, Inc.** (Warsaw, IN)

Family ID: **92715433**

Assignee: **WARSAW ORTHOPEDIC, INC (WARSAW, IN)**

Appl. No.: **19/219337**

Filed: **May 27, 2025**

Related U.S. Application Data

parent US continuation 18121396 20230314 parent-grant-document US 12336741 child US 19219337

Publication Classification

Int. Cl.: **A61B17/70** (20060101)

U.S. Cl.:

CPC **A61B17/705** (20130101);

Background/Summary

TECHNICAL FIELD

[0001] The present disclosure generally relates to medical devices for the treatment of musculoskeletal disorders, and more particularly to a surgical system and method for correction of a spine disorder.

BACKGROUND

[0002] Spinal pathologies and disorders such as scoliosis and other curvature abnormalities, kyphosis, degenerative disc disease, disc herniation, osteoporosis, spondylolisthesis, stenosis, tumor, and fracture may result from factors including trauma, disease and degenerative conditions caused by injury and aging. Spinal disorders typically result in symptoms including deformity, pain, nerve damage, and partial or complete loss of mobility.

[0003] Non-surgical treatments, such as medication, rehabilitation and exercise can be effective, however, may fail to relieve the symptoms associated with these disorders. Surgical treatment of these spinal disorders includes correction, fusion, fixation, discectomy, laminectomy and implantable prosthetics. Correction treatments used for positioning and alignment may employ implants, such as vertebral rods and bone screws for stabilization of a treated section of a spine. This disclosure describes an improvement over these prior technologies.

SUMMARY

[0004] In one embodiment, in accordance with the principles of the present disclosure, a surgical system comprises a spine rod connector having a first portion and a second portion. The first portion comprises opposite first and second end surfaces and opposite first and second side surfaces each extending from the first end surface to the second end surface. The first portion comprises opposite top and bottom surfaces. The end and side surfaces each extend from the top surface to the bottom surface. The first portion comprises a first aperture extending between and through the end surfaces and spaced apart holes extending through the top surface such that the holes are in communication with the first aperture. The second portion comprises opposite first and second end surfaces and opposite first and second side surfaces each extending from the first end surface of the second portion to the second end surface of the second portion. The second portion comprises opposite top and bottom surfaces, The end and side surfaces of the second portion each extend from the top surface of the second portion to the bottom surface of the second portion. The second portion comprises a second aperture extending between and through the side surfaces of the second portion and spaced apart holes extending through the top surface of the second portion such that the holes of the second portion are in communication with the second aperture.

[0005] In some embodiments, the first aperture defines a first longitudinal axis disposed in a first plane and the second aperture defines a second longitudinal axis disposed in a second plane that is proximal to the first plane. In some embodiments, the top surface of the second portion is planar from the first end surface of the second portion to the second end surface of the second portion. In some embodiments, the top surface of the second portion is positioned between the top surface of the first portion and the bottom surface of the first portion. In some embodiments, the bottom surface of the second portion is positioned between the top surface of the first portion and the bottom surface of the first portion. In some embodiments, the second portion comprises a cutout extending through the first end surface of the second portion and the top surface of the second portion, the cutout being aligned with one of the holes of the first portion. In some embodiments, the second portion comprises a cutout extending through the second end surface of the second portion and the top surface of the second portion, the cutout being aligned with one of the holes of the first portion. In some embodiments, the second portion comprises a first cutout extending through the first end surface of the second portion and the top surface of the second portion, the first cutout being aligned with one of the holes of the first portion; and the second portion comprises a second cutout extending through the second end surface of the second portion and the top surface of the second portion, the second cutout being aligned with another one of the holes of the first portion. In some embodiments, the first portion includes only the first aperture extending through the end surfaces of the first portion and the second portion includes only the second aperture extending through the side surfaces of the second portion. In some embodiments, the holes of the second portion are positioned between the holes of the first portion. In some embodiments, the first aperture extends perpendicular to the second aperture. In some embodiments, the first aperture is in communication with the second aperture. In some embodiments, the holes are each threaded for engagement with a threaded setscrew. In some embodiments, the second portion is monolithically formed with the first portion. In some embodiments, the first aperture is positioned between the holes of the second portion.

[0006] In some embodiments, a surgical system comprises the spine rod connector, a first spinal rod positioned in the first aperture and a second spinal rod positioned in the second aperture. In some embodiments, the first aperture and the first spinal rod each have a non-circular diameter to prevent rotation of the first spinal rod within the first aperture and the second aperture and the second spinal rod each have a non-circular diameter to prevent rotation of the second spinal rod within the second aperture. some embodiments, the surgical system further comprises setscrews disposed in the holes of the first portion to fix the first spinal rod relative to the spine rod connector; and setscrews disposed in the holes of the second portion to fix the second spinal rod relative to the spine rod connector.

[0007] In one embodiment, a spine rod connector comprises a first portion and a second portion. The first portion comprises opposite first and second end surfaces and opposite first and second side surfaces each extending from the first end surface to the second end surface. The first portion comprises opposite top and bottom surfaces. The end and side surfaces each extend from the top surface to the bottom surface. The first portion comprises a first aperture extending between and through the end surfaces and spaced apart holes extending through the top surface such that the holes are in communication with the first aperture. The second portion comprises opposite first and second end surfaces and opposite first and second side surfaces each extending from the first end surface of the second portion to the second end surface of the second portion. The second portion comprises opposite top and bottom surfaces. The end and side surfaces of the second portion each extend from the top surface of the second portion to the bottom surface of the second portion. The second portion comprises a second aperture extending between and through the side surfaces of the second portion and spaced apart holes extending through the top surface of the second portion such that the holes of the second portion are in communication with the second aperture. The first aperture defines a first longitudinal axis disposed in a first plane and the second aperture defines a second longitudinal axis disposed in a second plane that is proximal to the first plane. The top surface of the second portion is positioned between the top surface of the first portion and the bottom surface of the first portion. The bottom surface of the second portion is positioned between the top surface of the first portion and the bottom surface of the first portion. The second portion comprises a first cutout extending through the first end surface of the second portion and the top surface of the second portion, the first cutout being aligned with one of the holes of the first portion. The second portion comprises a second cutout extending through the second end surface of the second portion and the top surface of the second portion, the second cutout being aligned with another one of the holes of the first portion. The first portion includes only the first aperture extending through the end surfaces of the first portion and the second portion includes only the second aperture extending through the side surfaces of the second portion. The first aperture extends perpendicular to the second aperture. The first aperture is in communication with the second aperture. The second portion is monolithically formed with the first portion. The first aperture is positioned between the holes of the second portion.

[0008] In one embodiment, in accordance with the principles of the present disclosure, a surgical system comprises a spine rod connector comprising a first portion and a second portion. The first portion comprises opposite first and second end surfaces and opposite first and second side surfaces each extending from the first end surface to the second end surface. The first portion comprises opposite top and bottom surfaces. The end and side surfaces each extend from the top surface to the bottom surface. The first portion comprises a first aperture extending between and through the end surfaces and spaced apart holes extending through the top surface such that the holes are in communication with the first aperture. The second portion comprises opposite first and second end surfaces and opposite first and second side surfaces each extending from the first end surface of the second portion to the second end surface of the second portion. The second portion comprises opposite top and bottom surfaces. The end and side surfaces of the second portion each extend from the top surface of the second portion to the bottom surface of the second portion. The second portion comprises a second aperture extending between and through the side surfaces of the second portion and spaced apart holes extending through the top surface of the second portion such that the holes of the second portion are in communication with the second aperture. The system includes a first spinal rod positioned in the first aperture. The system includes a second spinal rod positioned in the second aperture. The system includes setscrews disposed in the holes of the first portion to fix the first spinal rod relative to the spine rod connector. The system includes setscrews disposed in the holes of the second portion to fix the second spinal rod relative to the spine rod connector. The first aperture and the first spinal rod each have a non-circular diameter to prevent rotation of the first spinal rod within the first aperture and the second aperture and the second spinal rod each have a non-circular diameter to prevent rotation of the second spinal rod within the second aperture. The first aperture defines a first longitudinal axis disposed in a first plane and the second aperture defines a second longitudinal axis disposed in a second plane that is proximal to the first plane. The top surface of the second portion is positioned between the top surface of the first portion and the bottom surface of the first portion. The bottom surface of the second portion is positioned between the top surface of the first portion and the bottom surface of the first portion. The second portion comprises a first cutout extending through the first end surface of the second portion and the top surface of the second portion, the first cutout being aligned with one of the holes of the first portion. The second portion comprises a second cutout extending through the second end surface of the second portion and the top surface of the second portion, the second cutout being aligned with another one of the holes of the first portion. The first portion includes only the first aperture extending through the end surfaces of the first portion and the second portion includes only the second aperture extending through the side surfaces of the second portion. The first aperture extends perpendicular to the second aperture. The first aperture is in communication with the second aperture. The second portion is monolithically formed with the first portion. The first aperture is positioned between the holes of the second portion.

[0009] Additional features and advantages of various embodiments will be set forth in part in the description that follows, and in part will be apparent from the description, or may be learned by practice of various embodiments. The objectives and other advantages of various embodiments will be realized and attained by means of the elements and combinations particularly pointed out in the description and appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The present disclosure will become more readily apparent from the specific description accompanied by the following drawings, in which:

[0011] FIG. **1** is a plan view of a spinal implant of a surgical system disposed with vertebrae, rods of the surgical system and a setscrew of the surgical system, in accordance with the principles of the present disclosure;

[0012] FIG. **2** is a perspective view of the spinal implant shown in FIG. **1**;

[0013] FIG. **3** is a perspective, cross-sectional view of the spinal implant shown in FIG. **1**;

[0014] FIG. **4** is a perspective view of the spinal implant shown in FIG. **1**;

[0015] FIG. **5** is a side, cross-sectional view of the spinal implant shown in FIG. **1**;

[0016] FIG. **6** is a plan view showing the rods shown in FIG. **1** disposed with the spinal implant shown in FIG. **1**;

[0017] FIG. **7** is a plan view showing the rods shown in FIG. **1** disposed with the spinal implant shown in FIG. **1** and the setscrew shown in FIG. **1** disposed with the spinal implant and one of the rods;

[0018] FIG. **8** is a perspective view of one embodiment of the spinal implant shown in FIG. **1**, in accordance with the principles of the present disclosure; and

[0019] FIG. **9** is a perspective view of one embodiment of the spinal implant shown in FIG. **1**, in accordance with the principles of the present disclosure.

[0020] Like reference numerals indicate similar parts throughout the figures.

DETAILED DESCRIPTION

[0021] In some embodiments, the surgical system of the present disclosure may be employed to treat spinal disorders such as, for example, degenerative disc disease, disc herniation, osteoporosis, spondylolisthesis, stenosis, scoliosis and other curvature abnormalities, kyphosis, tumor and fractures. In some embodiments, the surgical system of the present disclosure may be employed with other osteal and bone related applications, including those associated with diagnostics and therapeutics. In some embodiments, the disclosed surgical system may be alternatively employed in a surgical treatment with a patient in a prone or supine position, and/or employ various surgical approaches to the spine, including anterior, posterior, posterior mid-line, direct lateral, postero-lateral, and/or antero-lateral approaches, and in other body regions. The surgical system of the present disclosure may also be alternatively employed with procedures for treating the lumbar, cervical, thoracic, sacral and pelvic regions of a spinal column. The surgical system of the present disclosure may also be used on animals, bone models and other non-living substrates, such as, for example, in training, testing and demonstration.

[0022] The surgical system of the present disclosure may be understood more readily by reference to the following detailed description of the embodiments taken in connection with the accompanying drawing figures, which form a part of this disclosure. It is to be understood that this application is not limited to the specific devices, methods, conditions or parameters described and/or shown herein, and that the terminology used herein is for the purpose of describing particular embodiments by way of example only and is not intended to be limiting. In some embodiments, as used in the specification and including the appended claims, the singular forms “a,” “an,” and “the” include the plural, and reference to a particular numerical value includes at least that particular value, unless the context clearly dictates otherwise. Ranges may be expressed herein as from “about” or “approximately” one particular value and/or to “about” or “approximately” another particular value. When such a range is expressed, another embodiment includes from the one particular value and/or to the other particular value. Similarly, when values are expressed as approximations, by use of the antecedent “about,” it will be understood that the particular value forms another embodiment. It is also understood that all spatial references, such as, for example, horizontal, vertical, top, upper, lower, bottom, left and right, are for illustrative purposes only and can be varied within the scope of the disclosure. For example, the references "upper" and "lower" are relative and used only in the context to the other, and are not necessarily "superior" and "inferior".

[0023] As used in the specification and including the appended claims, "treating" or "treatment" of a disease or condition refers to performing a procedure that may include administering one or more drugs to a patient (human, normal or otherwise or other mammal), employing implantable devices, and/or employing instruments that treat the disease, such as, for example, microdiscectomy instruments used to remove portions bulging or herniated discs and/or bone spurs, in an effort to alleviate signs or symptoms of the disease or condition. Alleviation can occur prior to signs or symptoms of the disease or condition appearing, as well as after their appearance. Thus, treating or treatment includes preventing or prevention of disease or undesirable condition (e.g., preventing the disease from occurring in a patient, who may be predisposed to the disease but has not yet been diagnosed as having it). In addition, treating or treatment does not require complete alleviation of signs or symptoms, does not require a cure, and specifically includes procedures that have only a marginal effect on the patient. Treatment can include inhibiting the disease, e.g., arresting its development, or relieving the disease, e.g., causing regression of the disease. For example, treatment can include reducing acute or chronic inflammation; alleviating pain and mitigating and inducing re-growth of new ligament, bone and other tissues; as an adjunct in surgery; and/or any repair procedure. In some embodiments, as used in the specification and including the appended claims, the term "tissue" includes soft tissue, ligaments, tendons, cartilage and/or bone unless specifically referred to otherwise.

[0024] The following discussion includes a description of a surgical system including implants, related components and methods of employing the surgical system in accordance with the principles of the present disclosure. Alternate embodiments are also disclosed. Reference is made in detail to the exemplary embodiments of a surgical system **20**, which are illustrated in the accompanying figures.

[0025] The components of surgical system **20** can be fabricated from biologically acceptable materials suitable for medical applications, including metals, synthetic polymers, ceramics and bone material and/or their composites. For example, the components of surgical system **20**, individually or collectively, can be fabricated from materials such as stainless steel alloys, aluminum, commercially pure titanium, titanium alloys, Grade 5 titanium, super-elastic titanium alloys, cobalt-chrome alloys, superelastic metallic alloys (e.g., Nitinol, super elasto-plastic metals, such as GUM METAL®), ceramics and composites thereof such as calcium phosphate (e.g., SKELITE™), thermoplastics such as polyaryletherketone (PAEK) including polyetheretherketone (PEEK), polyetherketoneketone (PEKK) and polyetherketone (PEK), carbon-PEEK composites, PEEK-BaSO4 polymeric rubbers, polyethylene terephthalate (PET), fabric, silicone, polyurethane, silicone-polyurethane copolymers, polymeric rubbers, polyolefin rubbers, hydrogels, semi-rigid and rigid materials, elastomers, rubbers, thermoplastic elastomers, thermoset elastomers, elastomeric composites, rigid polymers including polyphenylene, polyamide, polyimide, polyetherimide, polyethylene, epoxy, bone material including autograft, allograft, xenograft or transgenic cortical and/or corticocancellous bone, and tissue growth or differentiation factors, partially resorbable materials, such as, for example, composites of metals and calcium-based ceramics, composites of PEEK and calcium based ceramics, composites of PEEK with resorbable polymers, totally resorbable materials, such as, for example, calcium based ceramics such as calcium phosphate, tri-calcium phosphate (TCP), hydroxyapatite (HA)-TCP, calcium sulfate, or other resorbable polymers such as polyaetide, polyglycolide, polytyrosine carbonate, polycaroplaetohe and their combinations.

[0026] Various components of surgical system **20** may have material composites, including the above materials, to achieve various desired characteristics such as strength, rigidity, elasticity, compliance, biomechanical performance, durability and radiolucency or imaging preference. The components of surgical system **20**, individually or collectively, may also be fabricated from a heterogeneous material such as a combination of two or more of the above-described materials. The components of surgical system **20** may be monolithically formed, integrally connected or include fastening elements and/or instruments, as described herein.

[0027] Surgical system **20** is employed, for example, with a fully open surgical procedure, a minimally invasive procedure including percutaneous techniques, and mini-open surgical techniques to deliver and introduce instrumentation and/or one or more spinal implants, such as, for example, one or more components of a bone fastener, at a surgical site of a patient, which includes, for example, a spine. In some embodiments, the spinal implant can include one or more components of one or more spinal constructs, such as, for example, interbody devices, interbody cages, bone fasteners, spinal rods, tethers, connectors, plates and/or bone graft, and can be employed with various surgical procedures including surgical treatment of a cervical, thoracic, lumbar and/or sacral region of a spine.

[0028] The surgical system disclose herein includes a spine rod connector. In some embodiments, the spine rod connector is configured for use in, among other things, sacrectomy surgery. In particular, during sacrectomy surgery, rods are attached to the iliac and lumbar spine. These rods can cross such that the rods extend at 90 degrees relative to one another. The rods may be posterior or anterior. The spine rod connector disclosed herein is configured easily facilitate the two crossing rods. For example, the spine rod connector is configured to address crossing rods in a compact versatile way so that the number of crossing rods or the location and orientation of the rods can be addressed easily. The ease of use and versatility of the spine rod connector offers surgeons a fast and robust solution to a complex surgery. In some embodiments, the spine rod connector includes rod apertures that are spaced apart in different planes and overlapping.

[0029] The surgical system disclose herein includes the spine rod connector and other components for use with the spine rod connector, such as, for example, a plurality of spinal rods and a plurality of setscrews, as discussed herein. Reference will now be made in detail to certain embodiments of the invention, examples of which are illustrated in the accompanying drawings. While the invention will be described in conjunction with the illustrated embodiments, it will be understood that they are not intended to limit the invention to those embodiments. On the contrary, the invention is intended to cover all alternatives, modifications, and equivalents that may be included within the invention as defined by the appended claims.

[0030] Surgical system **20** includes an implant, such as, for example, a spine rod connector **22**. Connector **22** includes a first portion **24** and a second portion **26**. In some embodiments, portion **24** and portion **26** are integrally and/or monolithically formed such that portion **26** cannot be removed from portion **24** without breaking portion **26** and/or portion **24**, and vice versa. In some embodiments, at least one of portion **24** and portion **26** is made from at least one of the materials discussed herein. For example, in some embodiments, portion **24** and/or portion **26** is/are made from a material that allows portion **24** and/or portion **26** to flex and/or bend without breaking portion **24** and/or portion **26** when stresses are applied to portion **24** and/or portion **26** by one or more spinal rods. In some embodiments, portion **24** and/or portion **26** is/are made from a rigid material that causes portion **24** and/or portion **26** to break when stresses are applied to portion **24** and/or portion **26** by one or more spinal rods.

[0031] Portion **24** comprises a first end surface **28** and an opposite end surface **30**. Portion **24** comprises a first side surface **32** and an opposite second side surface **34** each extending from surface **28** to surface **30**. Portion **24** comprises a top surface **36** and an opposite bottom surface **38**. In some embodiments, surface **38** extends parallel to surface **36**. Surfaces **30**, **32**, **34**, **36** each extend from surface **36** to surface **38**. Portion **24** comprises a first aperture **40** extending between and through surfaces **28**, **30** and spaced apart first and second holes **42**, **44** each extending through surface **36** such that holes **42**, **44** are each in communication with aperture **40**.

[0032] In some embodiments, portion **24** includes only aperture **40** extending through surfaces **28**,

30. That is, portion **24** does not include any aperture, channel, passageway, conduit, etc. that extends through surfaces **28**, **30** in addition to aperture **40**. In some embodiments, aperture **40** is disposed equidistant between surfaces **32**, **34** and/or equidistant between surfaces **36**, **38**. That is, a midpoint of a diameter of aperture **40** is disposed equidistant between surfaces **32**, **34** and/or equidistant between surfaces **36**, **38**. In some embodiments, aperture **40** may have various cross section configurations, such as, for example, circular, oval, oblong, triangular, rectangular, square, polygonal, irregular, uniform, non-uniform, variable, tubular and/or tapered. In some embodiments, aperture **40** has a uniform diameter along an entire length of aperture **40**. That is, aperture **40** has a uniform diameter from surface **28** to surface **30**. In some embodiments, holes **42**, **44** are each threaded for engagement with a coupling member, such as, for example, a setscrew such that the setscrews directly engage a spinal construct, such as, for example, a spinal rod positioned within aperture **40** to prevent movement of the spinal rod relative to connector **22**, as discussed herein. In some embodiments, surface **28** is entirely planar. That is, surface **28** is planar from surface **32** to surface **34** and/or from surface **36** to surface **38**. In some embodiments, surface **30** is entirely planar. That is, surface **30** is planar from surface **32** to surface **34** and/or from surface **36** to surface **38**. In some embodiments, surface **38** is entirely planar. That is, surface **38** is planar from surface **32** to surface **34** and/or from surface **28** to surface **30**. In some embodiments, holes **42**, **44** extend parallel to one another. In some embodiments, hole **42** and/or hole **44** extends perpendicular to aperture **40**. In some embodiments, hole **42** may be disposed at alternative orientations relative to hole **44** and/or at least one of holes **42**, **44** may be disposed at alternate orientations, relative to aperture **40**, such as, for example, transverse and/or other angular orientations such as acute or obtuse, co-axial and/or may be offset or staggered.

[0033] Portion **26** is coupled to portion **24** and comprises a first end surface **46** and an opposite end surface **48**. Portion **26** comprises a first side surface **50** and an opposite second side surface **52** each extending from surface **46** to surface **48**. Portion **26** comprises a top surface **54** and an opposite bottom surface **56**. In some embodiments, surface **56** extends parallel to surface **54**. In some embodiments, surface **54** and/or surface **56** extends parallel to surface **36** and/or surface **38**. Surfaces **46**, **48**, **50**, **52** each extend from surface **54** to surface **56**. Portion **26** comprises a second aperture **58** extending between and through surfaces **50**, **52** and spaced apart first and second holes **60**, **62** each extending through surface **54** such that holes **60**, **62** are each in communication with aperture **58**.

[0034] In some embodiments, channel **58** is positioned equidistant between surface **28** and surface **30** and/or channel **40** is positioned equidistant between surface **50** and surface **52**. In some embodiments, portion **26** includes only aperture **58** extending through surfaces **50**, **52**. That is, portion **26** does not include any aperture, channel, passageway, conduit, etc. that extends through surfaces **50**, **52** in addition to aperture **58**. In some embodiments, aperture **58** is disposed equidistant between surfaces **48**, **50** and/or equidistant between surfaces **54**, **56**. That is, a midpoint of a diameter of aperture **58** is disposed equidistant between surfaces **48**, **50** and/or equidistant between surfaces **54**, **56**. In some embodiments, aperture **58** may have various cross section configurations, such as, for example, circular, oval, oblong, triangular, rectangular, square, polygonal, irregular, uniform, non-uniform, variable, tubular and/or tapered. In some embodiments, aperture **58** has a uniform diameter along an entire length of aperture **58**. That is, aperture **58** has a uniform diameter from surface **50** to surface **52**. In some embodiments, holes **60**, **62** are each threaded for engagement with a coupling member, such as, for example, a setscrew such that the setscrews directly engage a spinal construct, such as, for example, a spinal rod positioned within aperture **58** to prevent movement of the spinal rod relative to connector **22**, as discussed herein. In some embodiments, holes **60**, **62** are positioned between hole **42** and hole **44**. In some embodiments, surface **50** is entirely planar. That is, surface **50** is planar from surface **46** to surface **48** and/or from surface **54** to surface **56**. In some embodiments, surface **52** is entirely planar. That is, surface **52** is planar from surface **46** to surface **48** and/or from surface **54** to surface **56**. In some embodiments, holes **60**, **62** extend parallel to one another. In some embodiments, hole **60** and/or hole **62** extends perpendicular to aperture **58**. In some embodiments, hole **60** may be disposed at alternative orientations relative to hole **62** and/or at least one of holes **60**, **62** may be disposed at alternate orientations, relative to aperture **58**, such as, for example, transverse and/or other angular orientations such as acute or obtuse, co-axial and/or may be offset or staggered.

[0035] In some embodiments, at least a portion of surface **54** is planar from surface **50** to surface **52**. That is, a portion of surface **54** between holes **60**, **62** and surface **46** is planar from surface **50** to surface **52** and a portion of surface **54** between holes **60**, **62** and surface **48** is planar from surface **50** to surface **52**. As such, a portion of surface **54** that hole **60** extends through is co-planar and/or level with a portion of surface **54** that hole **62** extends through. In some embodiments, at least a portion of surface **54** is planar from surface **46** to surface **48**. That is, at least a portion of surface **54** is planar from surface **50** to surface **52** and at least a portion of surface **54** is planar from surface **46** to surface **48**. In some embodiments, at least one of the portions of surface **54** that is planar from surface **50** to surface **52** intersects at least one of the portions of surface **54** that is planar from surface **46** to surface **48**.

[0036] Aperture **40** defines a longitudinal axis X**1** and aperture **58** defines a longitudinal axis X**2**. In some embodiments, axis X**1** is a central longitudinal axis and axis X**2** is a central longitudinal axis. In some embodiments, aperture **40** is entirely linear such that aperture **40** extends parallel to axis X**1** along an entire length of aperture **40**. In some embodiments, aperture **58** is entirely linear such that aperture **58** extends parallel to axis X**2** along an entire length of aperture **58**. In some embodiments, surfaces **32**, **34** extend parallel to one another and surfaces **46**, **48** extend parallel to one another such that surfaces **32**, **34** are disposed at an angle α relative to surfaces **46**, **48**. Aperture **58** is thus disposed at angle α relative to aperture **40** such that axis X**2** is disposed at angle α relative to axis X**1**. In some embodiments, angle α is between 0 and 180 degrees. In some embodiments, angle α is between 10 and 170 degrees. In some embodiments, angle α is between 20 and 160 degrees. In some embodiments, angle α is between 30 and 160 degrees. In some embodiments, angle α is between 40 and 150 degrees. In some embodiments, angle α is between 50 and 140 degrees. In some embodiments, angle α is between 60 and 130 degrees. In some embodiments, angle α is between 70 and 120 degrees. In some embodiments, angle α is between 80 and 110 degrees. In some embodiments, angle α is between 90 and 100 degrees. In some embodiments, angle α is 90 degrees. In some embodiments, disposing apertures **40**, **58** at angle α disposes a spinal rod, such as, for example, a first spinal rod **64** disposed in aperture **40** is disposed at angle α relative to a spinal rod, such as, for example, a second spinal rod **66** that is disposed in aperture **58**.

[0037] In some embodiments, aperture **40** and rod **64** each have a non-circular diameter, are textured or have a non-circular diameter and are textured to prevent rotation of rod **64** when rod **64** is positioned within aperture **40** and/or aperture **58** and rod **66** each have a non-circular diameter or are textured to prevent rotation of rod **66** when rod **66** is positioned within aperture **58**. That is, aperture **40** has a cross-sectional configuration that corresponds or matches that of rod **64** to prevent rotation of rod **64** within aperture **40** and/or aperture **58** has a cross-sectional configuration that corresponds or matches that of rod **66** to prevent rotation of rod **66** within aperture **58**. For example, in some embodiments, portion **24** includes an inner surface having a plurality of planar sections **74** and a plurality of arcuate sections **76** that define aperture **40**. As shown in FIG. **2**, one of sections **76** is positioned between adjacent sections **74**, and vice versa. Similarly, in some embodiments, portion **26** includes an inner surface having a plurality of planar sections **78** and a plurality of arcuate sections **80** that define aperture **58**. As shown in FIG. **4**, one of sections **80** is positioned between adjacent sections **78**, and vice versa. In some embodiments, rod **64** has a configuration that includes a size, shape and diameter and rod **66** has the same configuration as rod **64**. In such configurations, aperture **40** and aperture **58** each have a configuration that includes a size, shape and diameter and the configurations or rods **64**, **66** match and/or correspond to the configurations of apertures **40**, **58**.

[0038] In some embodiments, at least a portion of aperture **40**, such as, for example, MP**1** is disposed in a first plane P**1** and at least a portion of aperture **58**, such as, for example, MP**2** is disposed in a second plane P**2** that extends parallel to plane P**1**. Plane P**2** is disposed proximal to plane P**2**, as shown in FIG. **5**, for example. This allows a spinal rod, such as, for example, rod **64** that is disposed in aperture **40** to be positioned distal to a spinal rod, such as, for example, rod **66** that is disposed in aperture **58**. This allows a bottom or distal surface of rod **64** to be distal to a bottom or distal surface of rod **66** when rod **64** is disposed in aperture **40** and rod **66** is disposed in aperture **58**. Similarly, this allows a top or proximal surface of rod **66** to be proximal to a top or proximal surface of rod when rod **64** is disposed in aperture **40** and rod **66** is disposed in aperture **58**. In some embodiments, aperture **58** is in communication with aperture **40**. In particular, in some embodiments, aperture **58** intersects and/or overlaps a portion of aperture **40**, as shown in FIG. **5**. This allows spinal rod **66** to directly engage spinal rod **64** when rod **64** is disposed in aperture **40** and rod **66** is disposed in aperture **58**. In some embodiments, planes P**1**, P**2** extend parallel to axis X**1**, axis X**2**, aperture **40** and/or aperture **58**. In some embodiments, plane P**1** is positioned equidistant between surfaces **36**, **38** and/or extends parallel to surface **36** and/or surface **38**. In some embodiments, plane P**2** is positioned equidistant between surfaces **54**, **56** and/or extends parallel to surface **54** and/or surface **56**. In some embodiments, portion **26** is coupled to portion **24** such that surface **54** is positioned between surfaces **36**, **38** and surface **56** is positioned between surfaces **36**, **38**. This offset configuration between portion **24** and portion **26** allows plane P**2** to be positioned proximal to plane P**1** and hence for aperture **58** to be positioned proximal to aperture **40**. In some embodiments, surfaces **36**, **54** extend parallel to one another and surface **54** is spaced a first distance D**1** apart from surface **36**; and surfaces **38**, **56** extend parallel to one another and surface **56** is spaced a second distance D**2** apart from surface **38**, as shown in FIG. **4**. In some embodiments, distance D**2** is equal to distance D**1**. In some embodiments, distance D**2** is more or less than distance D**1**.

[0039] In some embodiments, portion **26** comprises a first cutout **68** extending through surface **48** and surface **54** such that cutout **68** is aligned with hole **44** and a second cutout **70** extending through surface **46** and surface **54** such that cutout **70** is aligned with hole **42**. In some embodiments, cutout **68** provides space and/or clearance for a coupling member, such as, for example, a setscrew **72** that is threaded with hole **44** such that setscrew **72** directly engages rod **64** when rod **64** is disposed in aperture **40** to fix rod **64** relative to connector **22**. Similarly, cutout **70** provides space and/or clearance for a coupling member, such as, for example, setscrew **72** that is threaded with hole **42** such that setscrew **72** directly engages rod **64** when rod **64** is disposed in aperture **40** to fix rod **64** relative to connector **22**. In some embodiments, cutouts **68**, **70** are concavely curved and/or include a continuous radius of curvature to correspond and/or match the configuration of an outer surface of setscrew **72**, for example. In some embodiments, cutouts **68**, **70** are each positioned equidistant between surfaces **44**, **46** and/or equidistant between surfaces **50**, **52**. In some embodiments, a surface that defines cutout **68** is continuous with a surface that defines hole **44** and/or a surface that defines cutout **70** is continuous with a surface that defines hole **42**.

[0040] In some embodiments, connector **22** is configured such that part of a first section **24***a* of portion **24** extends from a first side **84** of portion **26** and part of a second section **24***b* of portion **24** extends from an opposite second side **86** of portion **26**, as shown in FIG. **4**, for example. That is, surfaces **32**, **34**, **36** of section **24***a* each extend from surface **46** of portion **26** and surfaces **32**, **34**, **36** of section **24***b* each extend from surface **48** of portion **26**. Surface **36** of section **24***a* is spaced apart from surfaces **32**, **34**, **36** of section **24***b* by portion **26**. That is, surface **36** of section **24***a* is spaced apart from surface **36** of section **24***b* by surfaces **46**, **48** of portion **26**. Surface **38**, on the other hand, extends continuously from surface **28** to surface **30**.

[0041] In some embodiments, connector **22** is configured such that part of a first section **26***a* of portion **26** extends from a first side **88** of portion **24** and part of a second section **26***b* of portion **26** extends from an opposite second side **90** of portion **24**, as shown in FIG. **4**, for example. That is, surfaces **46**, **48**, **56** of section **26***a* each extend from surface **32** of portion **24** and surfaces **46**, **48**, **56** of section **26***b* each extend from surface **34** of portion **24**. Surface **56** of section **26***a* is spaced apart from surface **56** of section **26***b* by portion **24**. That is, surface **56** of section **26***a* is spaced apart from surface **56** of section **26***b* by surfaces **32**, **34** of portion **24**. Surface **54**, on the other hand, extends continuously from surface **50** to surface **52**.

[0042] In one embodiment, shown in FIGS. **1-7**, connector **22** is configured such that surface **54** covers aperture **58** and holes **60**, **62** extend through surface **54**. In such embodiments, rod **64** is configured to be side loaded into aperture **40** and/or rod **66** is configured to be side loaded into aperture **58** in the direction shown by arrow A in FIG. **2** or in the direction shown by arrow B in FIG. **2**. In the embodiment shown in FIGS. **1-7**, surface **54** and/or surface **56** cover(s) a portion of aperture **40**. In one embodiment, shown in FIG. **8**, surface **54** is spaced apart from aperture **58** and/or aperture **58** extends through surface **54** such that surfaces **46**, **48** define holes **60**, **62**. This allows rod **66** to be top loaded into aperture **58** in the direction shown by arrow C in FIG. **8** before, during, or after rod **64** is side loaded into aperture **40**. In one embodiment, shown in FIG. **9**, connector **22** is similar to the embodiment shown in FIG. **8** in that surface **54** is spaced apart from aperture **58** and/or aperture **58** extends through surface **54** such that surfaces **46**, **48** define holes **60**, **62**. However, in the embodiment shown in FIG. **9**, a portion of surface **56** directly above aperture **40** is removed and/or aperture **40** extends through surface **56**. This allows rod **64** to be top loaded into aperture **40** in the direction shown by arrow C in FIG. **8** and then for rod **66** to be top loaded into aperture **58** in the direction shown by arrow C in FIG. **8** after rod **64** is top loaded into aperture **40**.

[0043] In assembly, operation and use, system **20**, similar to the systems and methods described herein, can include a one or more bone fasteners **82** in addition to connector **22**, rod, **64**, rod **66** and setscrews **72** and is employed with a surgical procedure for treatment of a spinal disorder affecting a section of a spine of a patient, as discussed herein. Surgical system **20** is employed with a surgical procedure for treatment of a condition or injury of an affected section of the spine. In some embodiments, one or more of bone fasteners **82** are the same or similar to one or more of the bone fasteners recited in U.S. patent application Ser. No. 16/487,057, filed on Aug. 18, 2019, U.S. patent application Ser. No. 17/744,950, filed on May 16, 2022 and/or U.S. patent application Ser. No. 17/749,920, filed on May 16, 2022. These applications are hereby incorporated herein by reference, in their entireties.

[0044] Bone fasteners **82** and one or a plurality of spinal implants, for example, spinal rods **64**, **66** can be delivered or implanted as a pre-assembled device or can be assembled in situ. The components of system **20** may be completely or partially revised, removed or replaced.

[0045] In use, to treat a selected section of vertebrae V, as shown in FIGS. **1** and **7**, a medical practitioner obtains access to a surgical site including vertebrae V in any appropriate manner, such as through incision and retraction of tissues. In some embodiments, system **20** can be used in any existing surgical method or technique including open surgery, mini-open surgery, minimally invasive surgery and percutaneous surgical implantation, whereby vertebrae V is accessed through a mini-incision, or a sleeve that provides a protected passageway to the area. Once access to the surgical site is obtained, the particular surgical procedure can be performed for treating the spine disorder.

[0046] An incision is made in the body of a patient and a cutting instrument (not shown) creates a surgical pathway for implantation of components of system **20**. A preparation instrument (not shown) can be employed to prepare tissue surfaces of vertebrae V, as well as for aspiration and irrigation of a surgical region.

[0047] One or more bone fasteners **82** is/are fixed with tissue to engage with vertebrae V along one side of vertebrae V. One or more bone fasteners **82** is/are fixed with tissue on either side of sacrum S. Connector **22** is positioned adjacent to sacrum S. Spinal rod **64** is inserted through receivers of two of bone fasteners **82** and aperture **40** of connector **22**, as shown in FIG. **1**. Spinal rod **66** is inserted through receivers of two of bone fasteners **82** and aperture **58** of connector **22**, as also shown in FIG. **1**. In some embodiments, rod **64** extends through an entire length of aperture **40** such that rod **64** extends through surfaces **28**, **30** and/or rod **66** extends through an entire length of aperture **58** such that rod **66** extends through surfaces **50**, **52** while rod **64** extends through an entire length of aperture **40** such that rod **64** extends through surfaces **28**, **30**. In some embodiments, rod **66** directly engages rod **64** when rod **66** is disposed in aperture **58** and rod **64** is disposed in aperture **40**, due to the overlap/intersection of apertures **40**, **58** shown in FIG. **5**. In some embodiments, rod **64** and/or rod **66** is/are sized, shaped and dimensioned such that rod **66** is spaced apart from (does not directly engage) rod **64** when rod **66** is disposed in aperture **58** and rod **64** is disposed in aperture **40**. Setscrews **82** engage with a surgical instrument, such as, for example, a driver (not shown), which advances setscrews **82** into engagement with the receivers of fasteners **82** to attach spinal rods **64**, **66** with vertebrae V, sacrum S, etc. Setscrews **82** are threaded into engagement with holes **42**, **44**, as shown in FIG. **7**, such that setscrews **82** directly engage rod **64** to fix rod **64** relative to connector **22**. Setscrews **82** are threaded into engagement with holes **60**, **62**, as shown in FIG. **7**, such that setscrews **82** directly engage rod **66** to fix rod **66** relative to connector **22**.

[0048] In some embodiments, rods **64**, **66** are fixed with vertebrae V and/or sacrum S to stabilize vertebrae V and affect growth for a correction treatment to treat spine pathologies, as described herein. In some embodiments, one or more of bone fasteners **82** are fixed with tissue, which may include selected vertebrae, for example, cervical, thoracic, lumber, sacral and/or iliac bone, one or more portions of vertebrae, for example, lamina, pedicle, spinous process, transverse process, cancellous and/or cortical bone surfaces, and/or ribs. In some embodiments, one or all of the components of system **20** can be delivered or implanted as a pre-assembled device or can be assembled in situ, in a selected order of assembly or the order of assembly of the particular components of system **20** can be varied according to practitioner preference, patient anatomy or surgical procedure parameters.

[0049] Upon completion of the procedure, the surgical instruments, assemblies and non-implanted components of system **20** are removed from the surgical site and the incision is closed. One or more of the components of system **20** can be made of radiolucent materials such as polymers. Radiomarkers may be included for identification under x-ray, fluoroscopy, CT or other imaging techniques. In some embodiments, the use of surgical navigation, microsurgical and image guided technologies may be employed to access, view and repair spinal deterioration or damage, with the aid of system **20**.

[0050] In some embodiments, **20** includes an agent, which may be disposed, packed, coated or layered within, on or about the components and/or surfaces of system **20**. In some embodiments, the agent may include bone growth promoting material, such as, for example, bone graft to enhance fixation of the bone fasteners with vertebrae. In some embodiments, the agent may include one or a plurality of therapeutic agents and/or pharmacological agents for release, including sustained release, to treat, for example, pain, inflammation and degeneration.

[0051] It will be understood that various modifications may be made to the embodiments disclosed herein. Therefore, the above description should not be construed as limiting, but merely as exemplification of the various embodiments. Those skilled in the art will envision other modifications within the scope and spirit of the claims appended hereto.

Claims

1-20. (canceled)

21. A spine rod connector, comprising: a first portion comprising opposite first and second end surfaces and a first aperture extending between and through the end surfaces, the first portion comprising spaced apart holes that are each in communication with the first aperture; and a second portion comprising opposite first and second side surfaces and a second aperture extending between and through the side surfaces, the second portion comprising spaced apart holes that are each in communication with the second aperture, the second aperture being in communication with the first aperture.

22. The spine rod connector recited in claim 21, wherein the first portion defines a top surface of the spine rod connector and the second portion defines an opposite bottom surface of the spine rod connector.

23. The spine rod connector recited in claim 22, wherein the spine rod connector extends along a longitudinal axis from the top surface to the bottom surface, the first aperture being offset relative to the second aperture along the longitudinal axis.

24. The spine rod connector recited in claim 21, wherein the first aperture is disposed in a first plane and the second aperture is disposed in a second plane, the first plane being proximal to the second plane.

25. The spine rod connector recited in claim 24, wherein the first plane extends parallel to the second plane.

26. The spine rod connector recited in claim 21, wherein the spine rod connector includes spaced apart cutouts extending through the first portion that are aligned with the holes of the second portion.

27. The spine rod connector recited in claim 26, wherein the spine rod connector includes spaced apart cutouts extending through the second portion that are aligned with the holes of the first portion.

28. The spine rod connector recited in claim 21, wherein the second portion extends from a bottom surface of the first portion.

29. A spinal system, comprising: a spine rod connector comprising: a first portion comprising opposite first and second end surfaces and a first aperture extending between and through the end surfaces, the first portion comprising spaced apart holes that are each in communication with the first aperture, and a second portion comprising opposite first and second side surfaces and a second aperture extending between and through the side surfaces, the second portion comprising spaced apart holes that are each in communication with the second aperture, the second aperture being in communication with the first aperture; a first rod disposed in the first aperture; and a second rod disposed in the second aperture.

30. The spinal system recited in claim 29, further comprising: first screws positioned in the holes of the first portion such that the first screws engage the first rod; and second screws positioned in the holes of the second portion such that the second screws engage the second rod.

31. The spinal system recited in claim 30, the screws are threaded with the holes.

32. The spinal system recited in claim 29, the first portion defines a top surface of the spine rod connector and the second portion defines an opposite bottom surface of the spine rod connector.

33. The spinal system recited in claim 32, the spine rod connector extends along a longitudinal axis from the top surface to the bottom surface, the first aperture being offset relative to the second aperture along the longitudinal axis.

34. The spinal system recited in claim 29, the first aperture is disposed in a first plane and the second aperture is disposed in a second plane, the first plane being proximal to the second plane.

35. The spinal system recited in claim 29, the apertures each have a non-circular diameter.

36. The spinal system recited in claim 29, the first aperture has a cross-sectional configuration that matches that of the first rod to prevent rotation of the first rod within the first aperture.

37. The spinal system recited in claim 36, the second aperture has a cross-sectional configuration that matches that of the second rod to prevent rotation of the second rod within the second aperture.

38. The spinal system recited in claim 29, the first rod directly engages the second rod.

39. The spinal system recited in claim 29, the first aperture is disposed at an angle relative to the second aperture between 90 degrees and 100 degrees.

40. A spinal system, comprising: a spine rod connector comprising: a first portion comprising opposite first and second end surfaces and a first aperture extending between and through the end surfaces, the first portion comprising spaced apart holes that are each in communication with the first aperture, and a second portion comprising opposite first and second side surfaces and a second aperture extending between and through the side surfaces, the second portion comprising spaced apart holes that are each in communication with the second aperture, the second aperture being in communication with the first aperture; a first rod disposed in the first aperture; a second rod disposed in the second aperture such that the second rod directly engages the first rod; first screws positioned in the holes of the first portion such that the first screws engage the first rod; and second screws positioned in the holes of the second portion such that the second screws engage the second rod, wherein the first aperture is disposed in a first plane and the second aperture is disposed in a second plane, the first plane being proximal to the second plane.